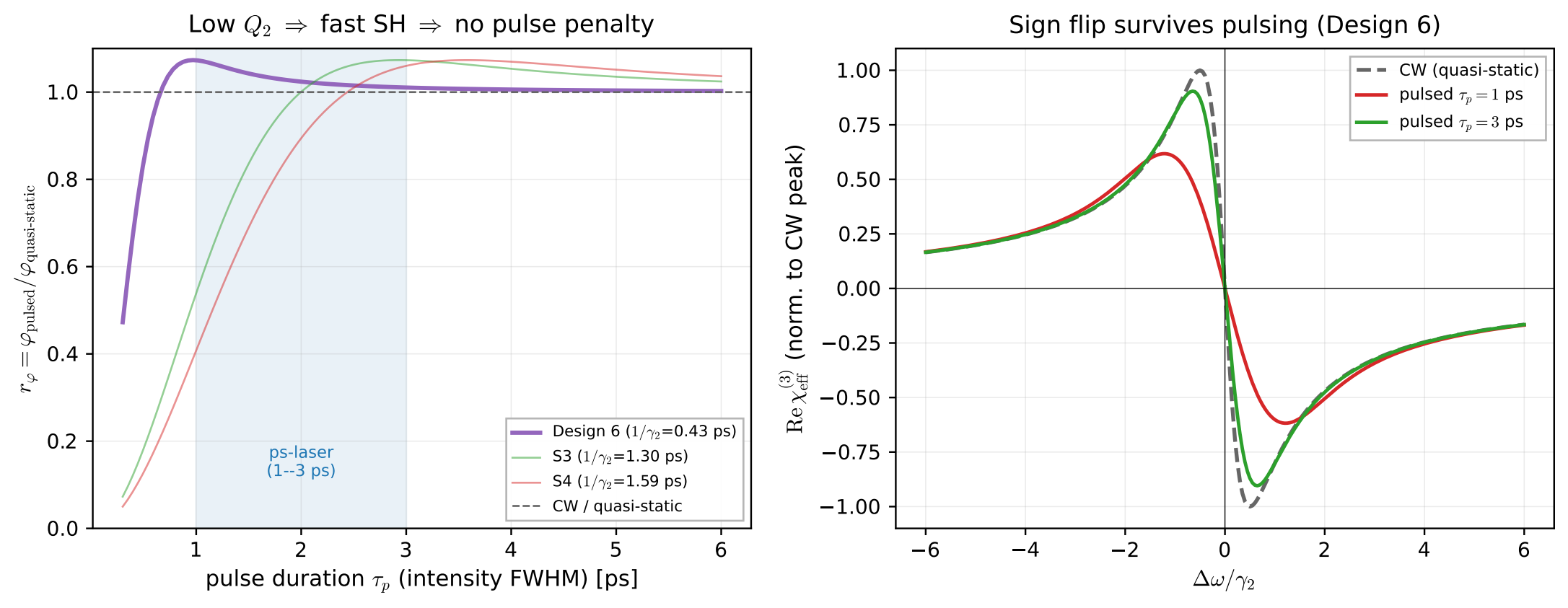
Low Q_2 ⇒ fast SH ⇒ no pulse penalty
$r_\varphi = \varphi_{\mathrm{pulsed}}/\varphi_{\mathrm{quasi\text{-}static}}$
pulse duration τ_p (intensity FWHM) [ps]
ps-laser
(1--3 ps)
Design 6 ($1/\gamma_2$=0.43 ps)
S3 ($1/\gamma_2$=1.30 ps)
S4 ($1/\gamma_2$=1.59 ps)
CW / quasi-static
Sign flip survives pulsing (Design 6)
$\mathrm{Re}\,\chi^{(3)}_{\mathrm{eff}}$ (norm. to CW peak)
$\Delta\omega/\gamma_2$
CW (quasi-static)
pulsed $\tau_p = 1$ ps
pulsed $\tau_p = 3$ ps